Updated 6DoF Sensing Framework: Scaling Limits and Spine Density

Analysis of 3mm Miniaturization Tax and Continuum Reconstruction

1 The "Miniaturization Tax" and Noise Amplification

Experimental modeling of the 3mm sensor board reveals a critical stability threshold. As the board diameter D decreases, the condition number $\kappa(\mathbf{M})$ grows exponentially.

- **Rotation Error (R_{err}):** Below 5mm, the leverage arm L is insufficient to distinguish tilt from translation. At 3mm, rotational noise is amplified by approximately $4\times$ compared to a 20mm baseline.
- **Translation Jitter:** While X, Y, Z are less dependent on L, they suffer from high-frequency "jagged" noise due to the low Signal-to-Noise Ratio (SNR) of small-aperture photodiodes.

2 Continuum Spine Reconstruction Density

To reconstruct the shape of a soft robot (e.g., "pool noodle" analogy) bending up to $\theta = 75°$ (1.31 rad) over a length $L_{spine} = 100$mm, we calculate the minimum bend radius ρ_{min}:

$$\rho_{min} = \frac{L_{spine}}{\theta_{rad}} \approx 76.3 \text{ mm} \tag{1}$$

Applying the Nyquist-criterion for curvature sampling, the sensor spacing S must satisfy:

$$S \leq \frac{\rho_{min}}{2} \approx 38 \text{ mm} \tag{2}$$

Recommended Configuration: To account for S-curves and complex strain localized at the 3mm board, a spacing of **20–25mm** is recommended, resulting in a 5-sensor "string of pearls" for a 100mm spine.

3 Mechanical Interaction: Rigid-Inclusion Correction

Because the 3mm PCB is rigid relative to the 5mm soft spine, the sensor creates a "dead zone" where the silicone cannot bend naturally. The effective sensing length L_{eff} is reduced by the inclusion diameter D:

$$L_{eff} = S - (1.5 \times D_{sensor}) \tag{3}$$

For a 3mm board, each sensor "blindfolds" approximately 4.5mm of the robot's local curvature.

```
% Calculate required sensors for a given max bend
L_spine = 100; % mm
max_bend_deg = 75;
rho_min = L_spine / deg2rad(max_bend_deg);
spacing = rho_min / 2; % Nyquist limit
num_sensors = ceil(L_spine / spacing);
fprintf('Required Sensors: %d at %0.1f mm intervals\n', num_sensors, spacing);
```

Listing 1: Optimal Density Calculation